

Taller Chat Cifrado

Por:

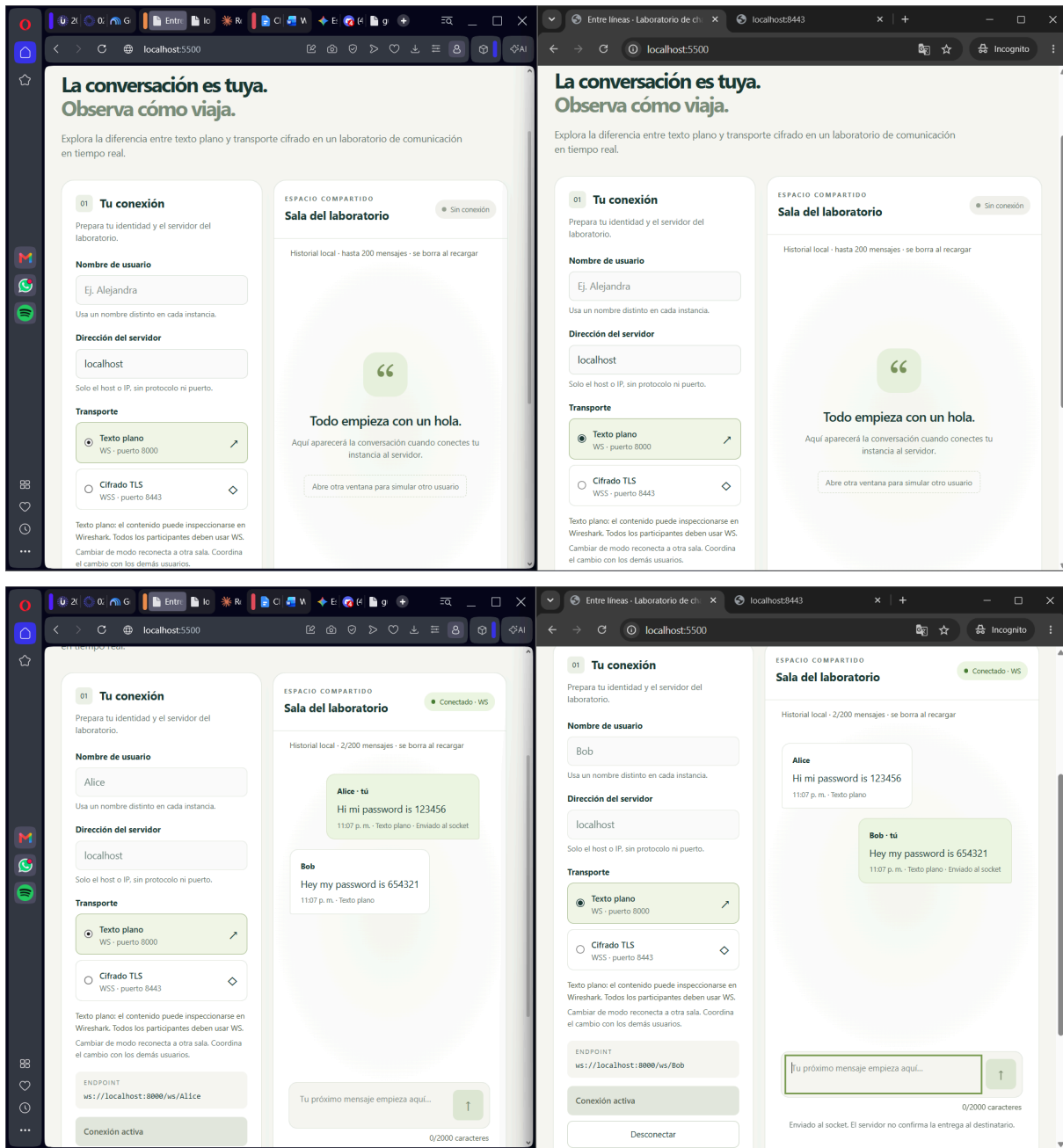
Santiago Zapata R (A00404790)

Juan David Salazar (A00404072)

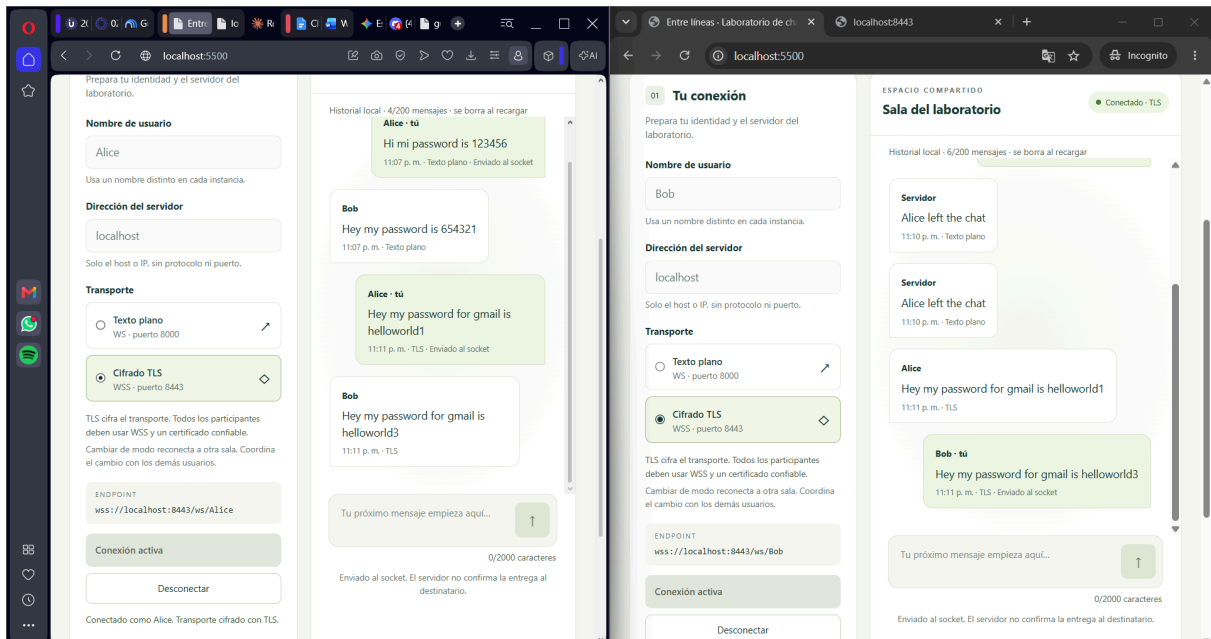
Samuel Lasso (A00404737)

Isaac Chaves(A00404410)

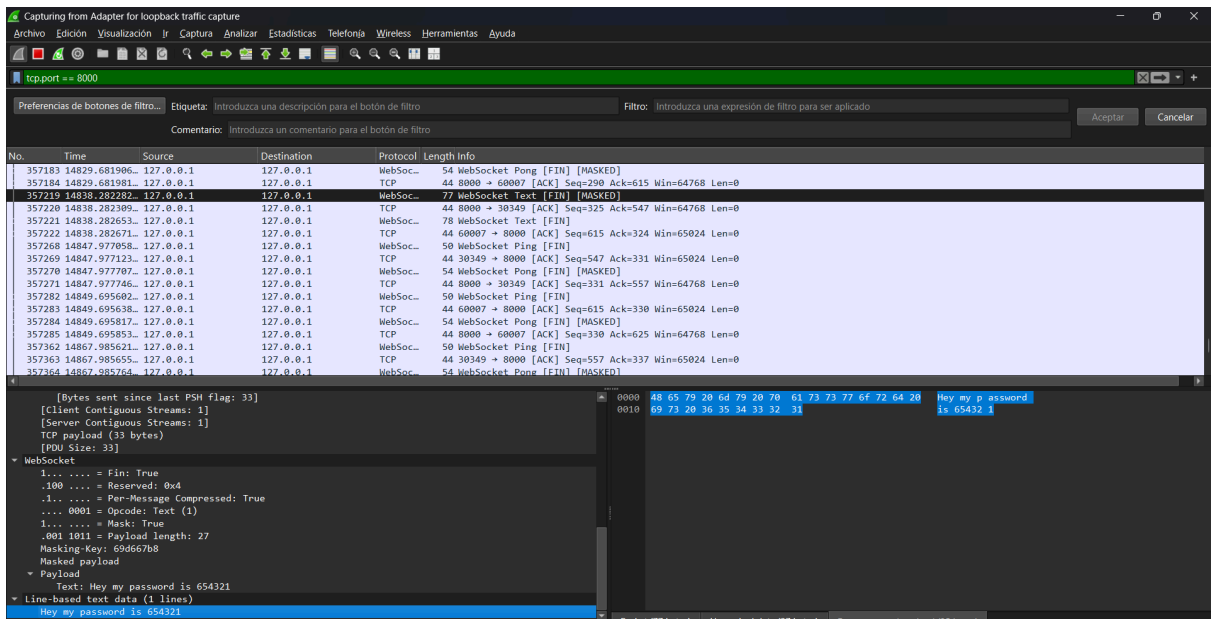
1. Chat funcionando en modo no cifrado

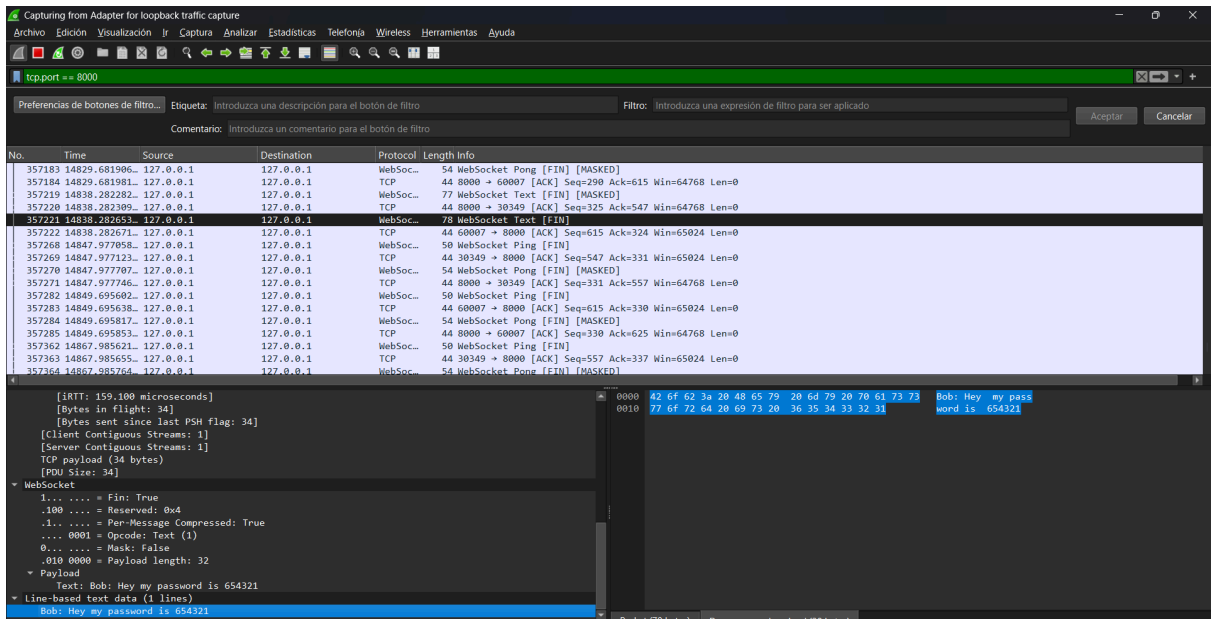


2. Chat funcionando en modo cifrado.

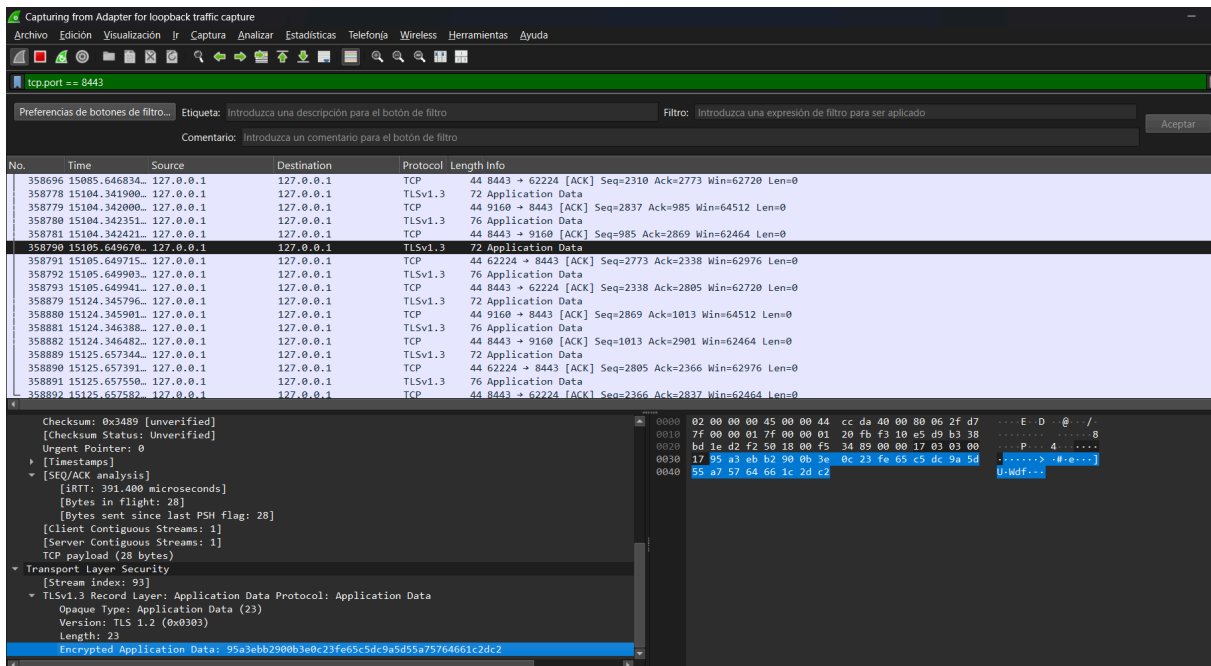


3. Captura de Wireshark del tráfico no cifrado.





4. Captura de Wireshark del tráfico cifrado.



Capturing from Adapter for loopback traffic capture

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

tcp.port == 8443

Preferencias de botones de filtro... Etiqueta: Introduzca una descripción para el botón de filtro Filtro: Introduzca una expresión de filtro para ser aplicado

Comentario: Introduzca un comentario para el botón de filtro

No.	Time	Source	Destination	Protocol	Length	Info
358696	15085.646834	127.0.0.1	127.0.0.1	TCP	44	8443 → 62224 [ACK] Seq=2310 Ack=2773 Win=62720 Len=0
358778	15104.341900	127.0.0.1	127.0.0.1	TLSv1.3	72	Application Data
358778	15104.342000	127.0.0.1	127.0.0.1	TCP	44	9160 → 8443 [ACK] Seq=2837 Ack=985 Win=64512 Len=0
358780	15104.342351	127.0.0.1	127.0.0.1	TLSv1.3	76	Application Data
358781	15104.342421	127.0.0.1	127.0.0.1	TCP	44	8443 → 9160 [ACK] Seq=985 Ack=2869 Win=62464 Len=0
358790	15105.649670	127.0.0.1	127.0.0.1	TLSv1.3	72	Application Data
358791	15105.649715	127.0.0.1	127.0.0.1	TCP	44	62224 → 8443 [ACK] Seq=2773 Ack=2338 Win=62976 Len=0
358792	15105.649803	127.0.0.1	127.0.0.1	TLSv1.3	76	Application Data
358793	15105.649941	127.0.0.1	127.0.0.1	TCP	44	8443 → 62224 [ACK] Seq=2338 Ack=2805 Win=62720 Len=0
358879	15124.345796	127.0.0.1	127.0.0.1	TLSv1.3	72	Application Data
358880	15124.345901	127.0.0.1	127.0.0.1	TCP	44	9160 → 8443 [ACK] Seq=2869 Ack=1013 Win=64512 Len=0
358881	15124.346388	127.0.0.1	127.0.0.1	TLSv1.3	76	Application Data
358882	15124.346482	127.0.0.1	127.0.0.1	TCP	44	8443 → 9160 [ACK] Seq=1013 Ack=2901 Win=62464 Len=0
358889	15125.657344	127.0.0.1	127.0.0.1	TLSv1.3	72	Application Data
358890	15125.657391	127.0.0.1	127.0.0.1	TCP	44	62224 → 8443 [ACK] Seq=2805 Ack=2366 Win=62976 Len=0
358891	15125.657550	127.0.0.1	127.0.0.1	TLSv1.3	76	Application Data
358892	15125.657582	127.0.0.1	127.0.0.1	TCP	44	8443 → 62224 [ACK] Seq=2366 Ack=2837 Win=62464 Len=0

Checksum: 0x906e [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
+ [SIO/ACK analysis]
+ [RTT: 391.400 microseconds]
+ [Bytes in flight: 32]
+ [Bytes sent since last PSH flag: 32]
+ [Client Contiguous Streams: 1]
+ [Server Contiguous Streams: 1]
+ TCP payload (32 bytes)
+ Transport Layer Security
+ [Stream index: 93]
+ TLSv1.3 Record Layer: Application Data Protocol: Application Data
+ Opaque Type: Application Data (23)
+ Version: TLS 1.2 (0x0303)
+ Length: 27
+ Encrypted Application Data: 4010ff5ec95bc65db842c1ebaa933ee7b175898e45e6a9a4534c46

0000 02 00 00 00 45 00 00 48 cc dc 40 00 80 06 2f d1 ... E · H · @ · · · /
0010 7f 00 00 01 7f 00 00 01 f3 10 20 fb bd 1e d2 f2 ... TP · · · n · · · · ·
0020 c5 d9 b3 54 50 18 00 f6 90 6e 00 00 17 03 03 00 ... · · · · ·
0030 1b 40 10 ff 5e c9 5b cc 5d b8 42 c1 6b aa 93 3a ... · · · · ·
0040 87 b1 75 89 8e 45 e6 a9 a4 53 4c 46 ... · · · · ·

5. Comparación de ambos escenarios.

Capturing from Adapter for loopback traffic capture

Archivo Edición Visualización Ir Captura Analizar Estadísticas Telefonía Wireless Herramientas Ayuda

tcp.port == 8000

Preferencias de botones de filtro... Etiqueta: Introduzca una descripción para el botón de filtro Filtro: Introduzca una expresión de filtro para ser aplicado

Comentario: Introduzca un comentario para el botón de filtro

No.	Time	Source	Destination	Protocol	Length	Info
357183	14829.681906	127.0.0.1	127.0.0.1	WebSoc...	54	WebSocket Pong [FIN] [MASKED]
357184	14829.681981	127.0.0.1	127.0.0.1	TCP	44	8000 → 60007 [ACK] Seq=290 Ack=615 Win=64768 Len=0
357219	14838.282282	127.0.0.1	127.0.0.1	WebSoc...	77	WebSocket Text [FIN] [MASKED]
357220	14838.282309	127.0.0.1	127.0.0.1	TCP	44	8000 → 30349 [ACK] Seq=325 Ack=547 Win=64768 Len=0
357221	14838.282653	127.0.0.1	127.0.0.1	WebSoc...	78	WebSocket Text [FIN]
357222	14838.282671	127.0.0.1	127.0.0.1	TCP	44	60007 → 8000 [ACK] Seq=615 Ack=324 Win=65024 Len=0
357268	14847.977058	127.0.0.1	127.0.0.1	WebSoc...	50	WebSocket Ping [FIN]
357269	14847.977123	127.0.0.1	127.0.0.1	TCP	44	30349 → 8000 [ACK] Seq=547 Ack=331 Win=65024 Len=0
357270	14847.977707	127.0.0.1	127.0.0.1	WebSoc...	54	WebSocket Pong [FIN] [MASKED]
357271	14847.977746	127.0.0.1	127.0.0.1	TCP	44	8000 → 30349 [ACK] Seq=331 Ack=557 Win=64768 Len=0
357282	14849.695602	127.0.0.1	127.0.0.1	WebSoc...	50	WebSocket Ping [FIN]
357283	14849.695638	127.0.0.1	127.0.0.1	TCP	44	60007 → 8000 [ACK] Seq=615 Ack=330 Win=65024 Len=0
357284	14849.695817	127.0.0.1	127.0.0.1	WebSoc...	54	WebSocket Pong [FIN] [MASKED]
357285	14849.695853	127.0.0.1	127.0.0.1	TCP	44	8000 → 60007 [ACK] Seq=330 Ack=625 Win=64768 Len=0
357362	14867.985621	127.0.0.1	127.0.0.1	WebSoc...	50	WebSocket Ping [FIN]
357363	14867.985655	127.0.0.1	127.0.0.1	TCP	44	30349 → 8000 [ACK] Seq=557 Ack=337 Win=65024 Len=0
357364	14867.985764	127.0.0.1	127.0.0.1	WebSoc...	54	WebSocket Pong [FIN] [MASKED]

[Bytes sent since last PSH flag: 33]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
+ TCP payload (33 bytes)
+ [PDU Size: 33]
+ WebSocket
+ 1... .. = Fin: True
+ 100 ... = Reserved: 0x4
+ 1... .. = Per-Message Compressed: True
+ ... 0001 = Opcode: Text (1)
+ 1... .. = Mask: True
+ .001 1011 = Payload length: 27
+ Masking-Key: 69d667b8
+ Masked payload
+ Payload
+ Text: Hey my password is 654321
+ Line-based text data (1 lines)
+ Hey my password is 654321

0000 48 65 79 20 6d 79 20 70 61 73 73 77 6f 72 64 20 Hey my p assword
0010 69 73 20 36 35 34 33 32 31 is 654321

Capturing from Adapter for loopback traffic capture

ArchivoEdiciónVisualizaciónIrCapturaAnalizarEstadísticasTelefoníaWirelessHerramientasAyuda

tcp.port == 8443

Preferencias de botones de filtro...

Etiqueta: Introduzca una descripción para el botón de filtro

Filtro: Introduzca una expresión de filtro para ser aplicado

Aceptar

Comentario: Introduzca un comentario para el botón de filtro

No.	Time	Source	Destination	Protocol	Length	Info
358696	15085.646834..	127.0.0.1	127.0.0.1	TCP	44	8443 → 62224 [ACK] Seq=2310 Ack=2773 Win=62720 Len=0
358778	15104.341900..	127.0.0.1	127.0.0.1	TLSv1.3	72	Application Data
358779	15104.342000..	127.0.0.1	127.0.0.1	TCP	44	9160 → 8443 [ACK] Seq=2837 Ack=985 Win=64512 Len=0
358780	15104.342351..	127.0.0.1	127.0.0.1	TLSv1.3	76	Application Data
358781	15104.342421..	127.0.0.1	127.0.0.1	TCP	44	8443 → 9160 [ACK] Seq=985 Ack=2869 Win=62464 Len=0
358790	15105.649670..	127.0.0.1	127.0.0.1	TLSv1.3	72	Application Data
358791	15105.649715..	127.0.0.1	127.0.0.1	TCP	44	62224 → 8443 [ACK] Seq=2773 Ack=2338 Win=62976 Len=0
358792	15105.649903..	127.0.0.1	127.0.0.1	TLSv1.3	76	Application Data
358793	15105.649941..	127.0.0.1	127.0.0.1	TCP	44	8443 → 62224 [ACK] Seq=2338 Ack=2805 Win=62720 Len=0
358879	15124.345796..	127.0.0.1	127.0.0.1	TLSv1.3	72	Application Data
358880	15124.345901..	127.0.0.1	127.0.0.1	TCP	44	9160 → 8443 [ACK] Seq=2869 Ack=1013 Win=64512 Len=0
358881	15124.346388..	127.0.0.1	127.0.0.1	TLSv1.3	76	Application Data
358882	15124.346482..	127.0.0.1	127.0.0.1	TCP	44	8443 → 9160 [ACK] Seq=1013 Ack=2901 Win=62464 Len=0
358889	15125.657344..	127.0.0.1	127.0.0.1	TLSv1.3	72	Application Data
358890	15125.657391..	127.0.0.1	127.0.0.1	TCP	44	62224 → 8443 [ACK] Seq=2805 Ack=2366 Win=62976 Len=0
358891	15125.657550..	127.0.0.1	127.0.0.1	TLSv1.3	76	Application Data
358892	15125.657582..	127.0.0.1	127.0.0.1	TCP	44	8443 → 62224 [ACK] Seq=2366 Ack=2837 Win=62464 Len=0

Checksum: 0x3489 [Unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
[Timestamps]
[SEQ/ACK analysis]
[iRTT: 391.400 microseconds]
[Bytes in flight: 28]
[Bytes sent since last PSH flag: 28]
[Client Contiguous Streams: 1]
[Server Contiguous Streams: 1]
TCP payload (28 bytes)

Transport Layer Security
[Stream index: 93]
TLSv1.3 Record Layer: Application Data Protocol: Application Data
Opaque Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 23
Encrypted Application Data: 95a3ebb2900b3e0c23fe65c5dc9a5d55a75764661c2dc2

0000 02 00 00 00 45 00 00 44 cc da 40 00 80 06 2f d7E D @ : /
0010 7f 00 00 01 7f 00 00 01 20 fb f3 10 e5 d9 b3 38 8
0020 bd 1e d2 f2 50 18 00 f5 34 89 00 00 17 03 03 00P...4.....
0030 17 95 a3 eb b2 90 0b 3e 0c 23 fe 65 c5 dc 9a 5d> .#e...
0040 55 a7 57 64 66 1c 2d c2 U-Wdf-...